

RILEY 10 WSW, OR, USA

WMO: 727010

Lat: 43.4711N

Lon: 119.6917W

Elev: 1397

StdP: 85.63

Time Zone: -8.00 (NAP)

Period: 03-19

WBAN: 04128

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.3	-14.5	-21.3	0.7	-17.9	-17.9	0.9	-12.8	7.1	2.2	6.3	1.5	0.9	N/A	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	33.4	14.0	31.9	13.6	30.2	13.1	15.1	29.2	14.4	28.7	13.8	27.9	2.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.4	16.4	9.1	8.5	15.9	8.0	7.9	15.4	46.7	29.1	44.8	28.9	42.9	27.8	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.6	4.9	DB	-23.1	36.2	2.9	1.5	-25.2	37.3	-26.9	38.2	-28.6	39.0	-30.7	40.2	(4)
(5)			WB	-23.7	16.7	2.8	1.0	-25.8	17.5	-27.4	18.1	-29.0	18.7	-31.1	19.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-3.1	-1.1	2.8	5.3	10.1	14.6	21.0	19.6	14.3	7.4	0.9	-3.7	(6)	
	DBStd	9.31	5.03	4.15	3.70	3.96	4.17	4.47	3.55	3.54	4.55	4.30	4.86	5.23	(7)	
	HDD10.0	1967	407	310	223	149	52	11	0	1	14	104	273	423	(8)	
	HDD18.3	4186	666	543	481	391	256	126	16	28	134	340	522	682	(9)	
	CDD10.0	1017	0	0	1	8	55	150	341	297	142	22	1	0	(10)	
	CDD18.3	193	0	0	0	0	1	15	99	67	11	0	0	0	(11)	
	CDH23.3	3606	0	0	0	4	66	364	1603	1165	385	19	0	0	(12)	
	CDH26.7	1455	0	0	0	0	11	114	740	473	114	3	0	0	(13)	
(14) Wind	WSAvg	2.2	1.8	2.1	2.4	2.6	2.5	2.5	2.4	2.3	2.1	2.0	1.9	1.8	(14)	
(15) Precipitation	PrecAvg	268	29	18	22	27	33	19	11	9	10	21	28	31	(15)	
	PrecMax	426	66	50	48	53	95	49	37	29	33	47	66	76	(16)	
	PrecMin	131	2	4	6	9	2	0	1	0	1	4	6	8	(17)	
	PrecStd	78	20	13	13	13	24	13	9	8	9	14	15	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	13.0	19.0	23.8	28.3	33.1	35.8	34.8	31.9	25.8	17.5	9.4	(19)	
		MCWB	3.8	5.9	7.9	10.1	12.6	14.9	15.6	13.9	13.0	10.9	8.4	4.8	(20)	
	2%	DB	7.1	9.9	15.7	20.8	25.1	29.9	34.0	32.8	29.7	22.2	14.0	7.2	(21)	
		MCWB	3.2	5.0	6.7	9.1	11.0	13.7	14.1	13.5	12.3	9.6	7.1	4.2	(22)	
	5%	DB	5.4	7.6	12.9	17.6	22.6	27.4	32.6	31.1	27.8	19.4	11.1	5.4	(23)	
		MCWB	2.3	3.5	5.5	7.5	10.4	12.6	13.8	13.2	11.9	8.8	5.9	2.9	(24)	
	10%	DB	3.8	5.6	10.4	14.6	20.0	25.1	31.1	29.5	25.5	16.7	8.6	3.8	(25)	
		MCWB	1.5	2.7	4.9	6.1	9.4	11.5	13.4	12.8	11.2	7.8	4.5	1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.2	9.0	10.8	13.3	16.4	16.4	15.2	14.7	12.2	9.7	6.1	(27)	
		MCDB	8.1	11.7	16.2	21.7	22.5	28.8	31.9	29.6	25.4	21.0	15.4	8.3	(28)	
	2%	WB	4.0	5.8	7.4	9.6	12.0	14.9	15.2	14.5	13.7	10.9	8.0	4.6	(29)	
		MCDB	6.7	9.6	12.9	19.3	21.6	28.1	29.7	29.3	26.4	20.1	13.2	6.5	(30)	
	5%	WB	2.8	4.4	6.3	8.0	11.0	13.6	14.6	13.9	12.7	9.7	6.7	3.2	(31)	
		MCDB	5.3	7.6	11.5	16.1	19.6	25.3	29.3	28.8	25.7	17.0	11.0	5.1	(32)	
	10%	WB	1.8	3.1	5.2	6.7	10.0	12.5	14.0	13.4	11.7	8.5	5.1	1.7	(33)	
		MCDB	4.1	5.6	10.0	13.3	18.1	23.4	28.7	27.9	24.6	14.7	8.5	3.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	12.3	14.4	14.9	16.3	18.1	18.0	17.9	15.4	12.0	9.8	(35)	
		MCDBR	11.8	12.6	17.4	20.6	19.7	19.6	19.1	19.7	21.5	20.1	15.5	10.4	(36)	
	5% WB	MCWBR	8.2	7.8	9.7	10.4	8.7	7.7	7.1	7.1	9.4	10.6	9.6	7.7	(37)	
		MCWBR	10.8	11.3	15.1	17.9	16.2	18.2	18.0	18.3	19.7	16.9	14.9	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.254	0.261	0.270	0.285	0.299	0.302	0.307	0.301	0.285	0.279	0.258	0.253	(40)	
		taud	2.593	2.619	2.618	2.564	2.540	2.551	2.545	2.555	2.587	2.601	2.616	2.587	(41)	
	Ebn at Noon		912	960	987	986	972	966	957	956	954	922	898	879	(42)	
		Edh at Noon	63	73	83	95	100	100	99	95	85	73	61	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.68	2.62	3.89	5.36	6.44	7.51	7.81	6.77	5.24	3.40	1.98	1.34	(44)	
	RadStd		0.25	0.32	0.33	0.46	0.47	0.37	0.28	0.32	0.28	0.27	0.18	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	N/A

Nomenclature: See separate page